◆ Chunk 1: DBMS Definition

Definition:
A Database Management System (DBMS) is software that manages databases and provides an interface to store, retrieve, and manipulate data.

Explanation:
It acts as a bridge between users/applications and the database, ensuring data is organized and accessible.

Key Points:

- Manages structured data
- Provides data security and integrity
- Supports multiple users

Example:
MySQL, Oracle, PostgreSQL

Interview Tip:
Mention "data management + interface + security".

◆ Chunk 2: Database

Definition:
A database is an organized collection of related data.

Explanation:
Data is stored in structured form for easy access and management.

Key Points:

- Stored electronically
- Organized into tables
- Easily retrievable

Example:
Student records database

Interview Tip:
Database = data, DBMS = software.

◆ Chunk 3: Types of DBMS

Definition:
Different models of organizing data.

Explanation:
Each model defines how data is stored and accessed.

Key Points:

- Hierarchical
- Network
- Relational
- NoSQL

Example:
Relational → MySQL

Interview Tip:
Relational DBMS is most widely used.

◆ Chunk 4: Relational Model

Definition:
Data is stored in tables (relations).

Explanation:
Tables consist of rows and columns with relationships.

Key Points:

- Row → Tuple
- Column → Attribute
- Table → Relation

Example:
Student(ID, Name)

Interview Tip:
Primary key uniquely identifies row.

◆ Chunk 5: Keys in DBMS

Definition:
Attributes used to identify records.

Explanation:
Keys ensure uniqueness and relationships.

Key Points:

- Primary Key
- Foreign Key
- Candidate Key
- Super Key

Example:
StudentID as Primary Key

Interview Tip:
Primary key = unique + not null.

🔹 Chunk 6: ER Model

Definition:
Entity-Relationship model represents data logically.

Explanation:
Used for database design before implementation.

Key Points:

- Entity
- Attribute
- Relationship

Example:
Student — enrolls — Course

Interview Tip:
ER diagram is frequently asked.

🔹 Chunk 7: Normalization

Definition:
Process of organizing data to reduce redundancy.

Explanation:
Breaks tables into smaller ones.

Key Points:

- 1NF, 2NF, 3NF
- Removes anomalies

Example:
Separate student and course tables

Interview Tip:
3NF is most commonly used.

🔹 **Chunk 8: Functional Dependency**

Definition:
Relationship between attributes.

Explanation:
One attribute determines another.

Key Points:

- X → Y
- Used in normalization

Example:
ID → Name

Interview Tip:
Basis for normalization.

🔹 **Chunk 9: SQL Basics**

Definition:
Structured Query Language for DB operations.

Explanation:
Used to interact with databases.

Key Points:

- DDL
- DML
- DCL

- TCL

Example:
SELECT * FROM Students;

Interview Tip:
SQL is case-insensitive.

🔹 **Chunk 10: DDL Commands**

Definition:
Data Definition Language commands.

Explanation:
Used to define database structure.

Key Points:

- CREATE
- ALTER
- DROP

Example:
CREATE TABLE Student(...);

Interview Tip:
DDL changes schema.

🔹 **Chunk 11: DML Commands**

Definition:
Data Manipulation Language commands.

Explanation:
Used to manipulate data.

Key Points:

- INSERT
- UPDATE
- DELETE

Example:
INSERT INTO Student VALUES(...);

Interview Tip:
DML modifies data only.

🔹 Chunk 12: Joins in SQL

Definition:
Combine data from multiple tables.

Explanation:
Used when data is distributed across tables.

Key Points:

- INNER JOIN
- LEFT JOIN
- RIGHT JOIN
- FULL JOIN

Example:
Student JOIN Course

Interview Tip:
INNER JOIN is most common.

🔹 Chunk 13: Transactions

Definition:
A sequence of operations as a single unit.

Explanation:
Ensures data consistency.

Key Points:

- Commit
- Rollback

Example:
Bank transfer

Interview Tip:
Transaction must be atomic.

🔹 Chunk 14: ACID Properties

Definition:
Properties ensuring reliable transactions.

Explanation:
Guarantees correctness of DB operations.

Key Points:

- Atomicity
- Consistency
- Isolation
- Durability

Example:
Money transfer consistency

Interview Tip:
Very frequently asked question.

🔹 Chunk 15: Indexing

Definition:
Technique to speed up data retrieval.

Explanation:
Works like a book index.

Key Points:

- Improves search speed
- Uses extra memory

Example:
Index on StudentID

Interview Tip:
Trade-off: speed vs space.

🔹 Chunk 16: Views

Definition:
Virtual tables.

Explanation:
Do not store data physically.

Key Points:

- Derived from queries
- Improves security

Example:
View of selected columns

Interview Tip:
Views don’t store actual data.

🔹 Chunk 17: Stored Procedures

Definition:
Predefined SQL code stored in DB.

Explanation:
Reusable and efficient.

Key Points:

- Improves performance
- Reduces redundancy

Example:
Procedure for inserting data

Interview Tip:
Used in enterprise systems.

🔹 Chunk 18: Triggers

Definition:
Automatically executed actions.

Explanation:
Triggered by events like insert/update.

Key Points:

- BEFORE/AFTER triggers
- Used for validation

Example:
Audit logs

Interview Tip:
Trigger runs automatically.

🔹 Chunk 19: Concurrency Control

Definition:
Managing multiple transactions.

Explanation:
Prevents conflicts.

Key Points:

- Locking
- Timestamping

Example:
Two users editing same data

Interview Tip:
Avoids data inconsistency.

🔹 Chunk 20: Deadlock in DBMS

Definition:
Transactions waiting indefinitely.

Explanation:
Occurs due to resource locking.

Key Points:

- Circular wait
- Detection required

Example:
T1 waits for T2, T2 waits for T1

Interview Tip:
Similar to OS deadlock.

🔹 Chunk 21: NoSQL Databases

Definition:
Non-relational databases.

Explanation:
Used for unstructured data.

Key Points:

- Key-value
- Document
- Column-based

Example:
MongoDB

Interview Tip:
Used in big data systems.

🔹 Chunk: Interview Questions

1. What is DBMS?
Software to manage databases.

2. DBMS vs File System?
DBMS provides security and consistency.

3. What is normalization?
Reducing redundancy.

4. What is a key?
Attribute to identify records.

5. Types of keys?
Primary, Foreign, Candidate.

6. What is SQL?
Query language for DB.

7. What are joins?
Combining tables.

8. What is ACID?
Transaction properties.

9. What is a transaction?
Unit of work.

10. What is indexing?
Faster data retrieval.

11. What is a view?
Virtual table.

12. What is a trigger?
Auto-executed action.

13. What is deadlock?
Transactions stuck waiting.

14. What is concurrency control?
Managing multiple transactions.

15. What is functional dependency?
Relation between attributes.

16. What is ER model?
Logical DB design.

17. What is DDL?
Schema definition.

18. What is DML?
Data manipulation.

19. What is NoSQL?
Non-relational DB.

20. What is stored procedure?
Reusable SQL code.